

Master's Thesis

Untersuchung von Top-Quark Paarproduktionen mit einem Higgs Zerfall in zwei W Bosonen in einfach-leptonischen Endzuständen unter Nutzung von ATLAS

Studies of top quark pair production associated with an Higgs boson decaying into two W bosons in single-leptonic final states using ATLAS

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1. Introduction

As far back as 400 BC, humans pondered about the fundamental makeup of the natural world. The earliest records of this go back to the Greek philosopher Democritus who proposed that matter cannot be infinitely divisible. Thus, there must exist the smallest, indivisible particle, which he termed the ‘atom’ (from Greek *atomos* meaning ‘indivisible’). Although this idea was not empirically tested but derived from philosophical thoughts, it marked the inception of the quest for a fundamental understanding of the universe.

Over the following millennia, this idea was refined by several researchers using different modern techniques. Today, research into the fundamental structure of nature continues with particle physics superseding nuclear physics. The contemporary aim of particle physics is to develop a comprehensive understanding of fundamental phenomena, centred around formulating and expanding upon the Standard Model of particle physics, which stands as the most successful description of the fundamental nature of the universe.

At the forefront of this quest for knowledge stands the Large Hadron Collider (LHC) situated at CERN (European Organization for Nuclear Research). The LHC, the world’s most powerful particle accelerator, collides protons which reach nearly the speed of light, mimicking conditions that prevailed in the universe mere moments after the Big Bang. Through detailed analysis of these collisions, scientists aim to validate and refine the Standard Model of particle physics.

One of the main experiments conducted at the LHC is the ATLAS experiment. Its detector is a huge toroidally shaped particle detector around the collision beam pipe and is built from vast arrays of detectors utilising cutting-edge technologies.

The task of this project is to reconstruct and analyse the Higgs-associated top quark pair production in which the Higgs boson decays into two W -bosons. In particular, a specific single-leptonic channel is targeted. Due to its challenging topology, this particular channel has not been analysed independently before. Hence, a modern approach is deployed utilising techniques such as neural networks and neutrino weighting. The goal is to fully reconstruct the targeted events from ATLAS datasets and separate these from other processes. A successful reconstruction with background separation would allow for some several measurements using this channel.

1. Introduction

This report begins by introducing the theoretical foundation of the Standard Model in Chapter 2. The following Chapter 3 introduces the experimental setup of the LHC and the ATLAS detector in particular. The techniques applied in this project are explained in Chapter 5 and Chapter 6, which cover *SPA-Net* and neutrino weighting, respectively. Chapter 4 covers the developed truth matching algorithm and the results of its validation. And finally, Chapter ?? summarises the current status of the project and includes a roadmap for the next steps.

2. The Standard Model of Particle Physics

The Standard Model of particle physics (SM) [1] stands as the most successful framework in understanding the fundamental particles and their interactions to date. It encompasses all known elementary particles along with their antiparticles, and three out of the four known fundamental forces: the strong, weak, and electromagnetic forces. Notably, gravity remains beyond the scope of the SM.

2.1. Gauge Symmetry and Spontaneous Symmetry Breaking

The SM is a renormalisable quantum field theory [2] characterised by an internal gauge symmetry that is denoted as a $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$. A gauge symmetry is a mathematical symmetry that describes how the Lagrangian is invariant under local transformations. This means that the laws of physics do not change under these transformations. Gauge transformations form a Lie group [3] which necessitates the generation of an underlying gauge field [4].

The $SU(3)_C$ group symmetry describes the strong interaction, stemming from quantum chromodynamics (QCD) [5–9]. It introduces colour charges and their corresponding mediators, the gluons, which are explained in more detail in Section 2.2.

The combined $SU(2)_L \otimes U(1)_Y$ group symmetry corresponds to the electromagnetic and to the weak force [10–14]. They introduce the electrical charge Q and the weak isospin T , respectively. Those two forces further combine at high energies on the order of 10^2 GeV via the electroweak unification. The latter entails the unification of the electromagnetic force [10–17], originating from quantum electrodynamics (QED), and the weak force, stemming from quantum flavour dynamics (QFD). The charge of the combined interaction is the weak hypercharge Y . However, the bosons $W^{0,1,2}$ and B^0 predicted by this symmetry are massless which is contradicting the massive bosons observed. This requires a process that introduces massive bosons.

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The Brout-Englert-Higgs Mechanism [18–20] explains how bosons, and fermions as well, acquire their mass. The theory introduces a quantum Higgs-field. Every particle interacting with the field becomes massive. At very high energies, this Higgs-field is symmetric. However, at energies which we observe under normal conditions, this symmetry is spontaneously broken [21]. One of the results of this is a non-zero vacuum expectation value. This vacuum energy introduces additional terms to the Lagrangian, spoiling the electroweak symmetry. These additional terms mix the four massless bosons of the theoretical electroweak symmetry (W^1, W^2, W^3, B^0) to generate the massive bosons $W^\pm, Z^0$ and the massless boson γ . The mass originates from the introduced interactions between the $W^{1,2}$ bosons and the Higgs-field. The W -bosons are associated with the $SU(2)_L$ symmetry group. The remaining two bosons are associated with the $U(1)_Y$ symmetry group.

Furthermore, one of the degrees of freedom introduced by the Higgs-field is not defined via the Brout-Englert-Higgs mechanism. It manifests as the scalar Higgs boson H [18].

2.2. Particles

The SM describes particles as elemental units of matter or forces. All particles have various properties such as mass, electrical charge and spin. Furthermore, each particle has an antiparticle that has the same properties as the complementary particle, but its electrical charge is inverted. These antiparticles are typically denoted with a bar on top of the particle identifier. Uncharged particles do not have a complementary antiparticle. At its core, the SM categorizes particles into two main groups by their spin. These groups are fermions and bosons. In the following the focus will be on particles, since antiparticles have identical properties but with an inverted electrical charge.

Fermions

In the SM, fermions are one of the two fundamental classes of elementary particles, the other being bosons. Fermions are particles with half-integer spin. They are the basic constituents of matter and obey the Fermi-Dirac statistics as well as the Pauli exclusion principle. Moreover, Fermions can be subdivided into quarks and leptons as further described in the following sections.

Quarks

Quarks are elementary particles that interact via the strong nuclear force, as well as the weak and electromagnetic forces which were introduced in Section 2.1. Quarks are

classified into three generations. In each generation, there is an up-type and a down-type quark defined by their electrical charge and weak isospin. The up-type quarks have a positive electrical charge of $+2/3$ times the elementary charge e and a weak isospin of $+1/2$. The down-type quarks have an electrical charge of $-1/3$ e and a weak isospin of $-1/2$.

For each generation, a weak isospin doublet can be defined containing two out of six quark flavours. Each quark flavour has unique properties. The first generation's doublet contains the up (u) and down (d') quark. The second generation consists of the charm (c) and strange (s') and the final and third generation includes the top (t) and bottom (b') quark.

Note that down-type quarks denoted as q' in the isospin doublets describe the weak eigenstate while q describes the mass eigenstate. These two eigenstates are related via the unitary Cabibbo-Kobayashi-Maskawa (CKM) matrix [22, 23]. The matrix describes the probability amplitude V_{ij} for the transition from quark flavour i to quark flavour j under the weak interaction. Furthermore, it describes the relation of the weak and mass eigenstates of down-type quarks

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}. \quad (2.1)$$

The entries in the CKM matrix are not predicted by theory. Hence, the amplitudes must be measured experimentally. The current best fitted values [24] are the following:

$$\begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} = \begin{bmatrix} 0.97373 \pm 0.00031 & 0.2243 \pm 0.00243 & 0.00382 \pm 0.00020 \\ 0.221 \pm 0.021 & 0.975 \pm 0.006 & 0.0408 \pm 0.0014 \\ 0.0086 \pm 0.0002 & 0.0415 \pm 0.0009 & 1.014 \pm 0.029 \end{bmatrix} \quad (2.2)$$

With increasing quark generation, the masses of the quarks also increases. The current best measurements of the masses for the up-type quarks and the down-type mass eigenstates [24] yield

$$\begin{aligned} m_u &= 2.16^{+0.49}_{-0.26} \text{ MeV}, & m_d &= 24.68^{+0.48}_{-0.17} \text{ MeV}, \\ m_c &= 1.27^{+0.02}_{-0.02} \text{ GeV}, & m_s &= 93.4^{+8.6}_{-3.4} \text{ MeV}, \\ m_t &= 172.69^{+0.30}_{-0.30} \text{ GeV}, & m_b &= 4.18^{+0.03}_{-0.02} \text{ GeV}. \end{aligned} \quad (2.3)$$

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Quarks also carry a property known as colour charge [5, 6]. It is analogous to the electric charge but associated with the strong nuclear force. However, unlike electric charge, which comes in positive and negative values, colour has six possible states: red, green, blue and their complements anti-red, anti-green, and anti-blue. Quarks always combine to a state in which all three colours are combined, or any colour combined with its complement [6]. These states are called colourless. This is known as colour confinement and is responsible for the fact that isolated quarks are never observed in nature.

The Top Quark

The top quark will be explained in more detail since its of particular interest for this project. The discovery of the top quark took place in 1995 at FERMILAB through the efforts of the DØ [25] and CDF [26] experiments that both discovered the top quark independently. The top quark is acknowledged as the most massive among all quarks. Producing top quarks needs significant energies owing to high mass of the top quark. Such energies are attainable in hadron colliders. The top quark decays before it hadronises due to its brief lifetime is approximately $\tau_t \approx 5 \cdot 10^{-25}$ s [24]. It decays into a W -boson and a down-type quark which then undergoes hadronisation.

The likelihood of decaying into a specific down-type quark is determined by the previously mentioned CKM matrix in Equation 2.2. Comparing the values of V_{td} and V_{ts} to V_{tb} from Equation 2.2 shows the predominant top decay mode to b quarks.

The W -boson, the second particle from the top decay, also decays further. As explained later, it can decay either into a charged lepton & neutrino pair or into a quark and antiquark pair. Thus, the resulting final state from a top decay include one quark and a lepton or three quark, depending on the W -boson decay mode.

b tagging

To precisely identify t quarks, an accurate algorithm to distinguish b is necessary. The reason for this is the top quarks predominant decay mode into b quarks which was explained previously. To achieve this, b tagging is introduced [27, 28].

The accurate identification of works by exploiting the properties of the b quark. It has a relatively large mass and a long lifetime. The latter is explained by its suppressed decay modes. Equations 2.2 shows that the V_{tb} element is the highest values for b quark row. Thus, the b flavour couples predominantly to the t flavour and decay modes into other flavours are suppressed. However, the t is significantly heavier than the b quark as seen in Equation 2.3. Hence, the decay into a top quark is also strongly suppressed.

Due to the long lifetime of the b quark, it can travel a measurable distance before decaying. Thus, its decay products do not originate from the primary vertex of the collision. Instead, reconstructing these decay products yields a displaced vertex position which is typically referred as ‘secondary vertex’. This reconstruction combined with the hard momentum spectra of the b quarks decay products yields a sufficiently accurate b tagging which reaches an efficiency of 70% [29] at the ATLAS experiment. This approach can also be applied, to a lesser extent, to charm tagging.

Leptons

Leptons are the second group of fermions. They do not experience the strong nuclear force and interact via the weak and electromagnetic forces. Analogue to the quarks, there are three generations of leptons: electron, muon and tau [1]. These are typically denoted as e , μ and τ , respectively.

Each generation consists of a weak isospin doublet containing an uncharged neutrino ν_l and its respective charged lepton l [1]. Charged leptons have an electrical charge of $-1e$ and a weak isospin of $-1/2$. Furthermore, the charged leptons are massive. Their masses increase with higher generations as current measurements [24] show

$$\begin{aligned} m_e &= 0.511 \pm 0.001 \text{ MeV}, \\ m_\mu &= 105.66 \pm 0.001 \text{ MeV}, \\ m_\tau &= 1776.86 \pm 0.12 \text{ MeV}. \end{aligned} \tag{2.4}$$

On the contrary, the uncharged neutrinos have no electrical charge [24], a weak isospin of $+1/2$ and are predicted to be massless by the SM [1]. However, the later prediction is conflicting with some observed phenomena that are explained in Section 2.3.

Bosons

Bosons are the second group of particles in the SM. Unlike the Fermions, bosons have integer values of spin and obey Bose-Einstein statistics. Bosons can be subdivided into gauge bosons that mediate three of the four fundamental forces and the Higgs boson.

Gauge Bosons

In the SM there are 4 different gauge bosons types [1] that are responsible for the interaction between particles of the SM. While many of their properties differ, they all have

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spin 1. The remaining properties are discussed in the following paragraphs.

The photon (γ) is the massless mediator of the electromagnetic force [11, 13, 14]. It itself is not electrical charged and thus, couples to positively and negatively charged particles equally. The behaviour of photons is determined by QED which was introduced in Section 2.1.

A gluon (g) is a type of massless gauge bosons that is responsible for the mediation of the strong nuclear force [5]. They couple to the colour charge of quarks. And unlike the photon, gluons themselves also carry colour charges which are exchange during a strong interaction. There are eight different possible colour combination for gluons due to the rules of QCD. Since they carry the charge they couple to, gluons are able to couple to other gluons. These processes are called self-couplings.

Lastly, there are $W^\pm$ and Z^0 bosons. These are the mediators of the weak nuclear force. In contrast to the other gauge bosons, these are massive due to symmetry breaking in the Brout-Englert-Higgs Mechanism which is explained in more detail in Section 2.1. Their masses were measured to be [24]

$$\begin{aligned} m_W &= 80.377 \pm 0.012 \text{ GeV}, \\ m_Z &= 91.1876 \pm 0.0021 \text{ GeV}. \end{aligned} \tag{2.5}$$

$W^\pm$ bosons have an electrical charge of $\pm 1e$. Moreover, they have a weak isospin. Its third component of the weak isospin is ± 1 , respectively. Fermions interacting with the W -bosons undergo a flavour transition. Charged leptons convert to lepton neutrinos and vice versa. Quarks also change their flavour. Their transition rate from one quark to the other is determined by the previously discussed CKM matrix. Couplings with the W -bosons are CP violating. The Z^0 -boson, as indicated by its superscript, has no electrical charge and its third component of the weak isospin is 0.

Higgs Boson

The Higgs boson (H^0) was the last particle of the SM discovered in 2012 by the combined efforts of the ATLAS [30] and CMS [31] experiments. The Higgs boson's detection gave strong support to the Brout-Englert-Higgs mechanism discussed in Section 2.1. It is electrically neutral and has spin 0, making it the only scalar boson. Moreover, it is the second-heaviest particle ever detected with current measurements [24] showing a mass of

$$m_H = 125.25 \pm 0.17 \text{ GeV}. \quad (2.6)$$

Furthermore, it represents the quantum manifestation of the Higgs field and thus, couples to mass of particles. Its production and decay branching ratios are strongly dependent on the mass of the interacting particle. The higher the mass, the stronger is the Higgs coupling resulting in higher branching ratios. The reason for this is the linearity between the Yukawa coupling strength and the (fermion) masses. Hence, the most common production modes are in combination with either t quarks or W -/ Z -bosons.

For the decay, the most likely modes are decays into pairs of b quarks or W - and Z -bosons. While the Higgs coupling of the t is stronger than the b , the decay into a t quark pair is strongly suppressed, due to its significantly higher mass $2m_t > m_H$. Couplings with the Higgs field (thus also with the H -boson) are described as Yukawa couplings that also generate the mass of fermions and bosons [18, 20]. Since the H -boson is massive, it can couple to itself, similar to the gluon. This gives rise to Higgs self-coupling.

2.3. Beyond the Standard Model

Despite the incredible range of phenomena that the SM can predict and explain for example the existence and properties of the W -, Z - and H -bosons as well as its precision, certain phenomena contradict its fundamental explanations.

Neutrino Oscillation

One of these problematic observations is neutrino oscillation [32, 33]. It describes how neutrinos change their flavour while propagating through space. The first indication was found by the Homestake Experiment in 1968 [34]. It detected solar neutrinos using a large tank of fluid and observed a neutrino deficit. In 1998, the Super-Kamiokande experiment studied atmospheric neutrinos and discovered a changing electron/muon neutrinos flux during the runtime [35].

The oscillation originates from the separation of the flavour eigenstates of the neutrinos (ν_e, ν_μ, ν_τ) and its mass eigenstates (ν_1, ν_2, ν_3). If these eigenstates are not equal, then neutrinos could change their flavour. The probability of a neutrino of a given flavour transitioning into another flavour depends on the mixing angles and mass differences between the mass eigenstates. The observed oscillation necessitates at least two massive neutrinos. Contradictory, the SM predicts all neutrinos as massless which would result in

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their flavour and mass eigenstates to be identical. Consequently, the oscillation would be prohibited and thus, the SM fails to account for neutrino oscillations.

Gravity

Another obvious problem with the SM is the absence of gravity. While the force of gravity is observed, the SM has no boson responsible for mediating gravity. Possible solutions include the introduction of a hypothetical massless and uncharged graviton [36] or string theory [37]. Without an explanation of gravity, the SM can not achieve to be a unified theory of everything.

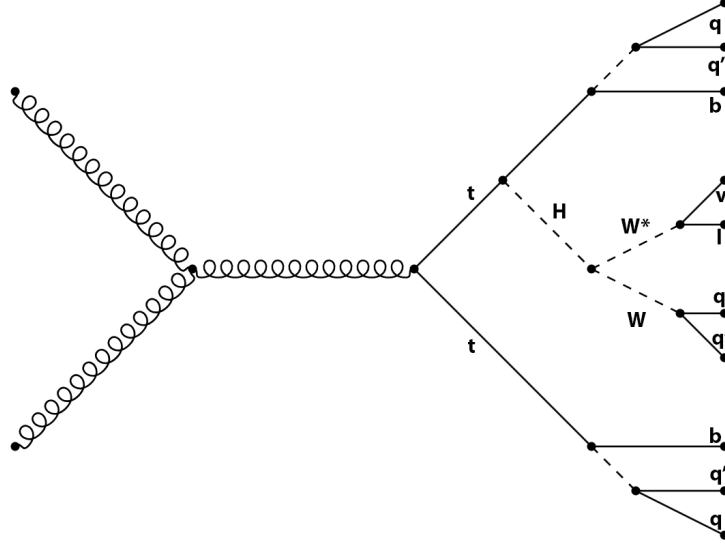
Further Contradictions

There are more contradicting observations that need explanations beyond the SM. These examples contain, but are not limited to, dark matter [38–40], a type of invisible mass needed for stellar observations, prediction of vacuum energy [41], a deviation between QFT and cosmological predictions and the hierarchy problem [42], a relatively low H -boson mass that necessitates fine-tuning. For each of the problems, several BSM explanations such as supersymmetry [43] or modified gravity [44] are possible.

2.4. Higgs-Associated Top Quark Pair Production

The targeted event topology is the Higgs-associated top quark pair production in which the Higgs boson decays into two W -bosons. The topology is a fully hadronic decaying $t\bar{t}$ pair with an additional H -boson being radiated. The latter decays to two W -bosons which then further decay semi-leptonically, which means one W -boson decays hadronically into two quarks while the other decays leptonically into a lepton neutrino pair. In the following, these events will be referred to as semileptonic $t\bar{t}(H \rightarrow WW^*)$ events. The corresponding Feynman diagram of the targeted topology can be seen in Figure 2.1.

Due to the insufficient mass of the H -boson to create two on-shell W -bosons, one of the two W -bosons must be off-shell. This is denoted by the asterisk superscript W^* . The assumption that the on-shell W -boson is decaying hadronically simplifies its reconstruction since the two selected decay jets should match the SM W -boson mass as close as possible. Thus, it can be used to suppress some of the main backgrounds ($t\bar{t} + \text{jets}$, $W + \text{jets}$) due to the constrain on the decay jet properties. However, it also necessitates that the off-shell W^* -boson decays leptonically which complicates the leptonic reconstruction. This will be solved by introducing a *Neutrino Weighting* technique.


 Figure 2.1.: Feynman diagram of the targeted $t\bar{t}(H \rightarrow WW^*)$ event topology.

The rarity of the semileptonic $t\bar{t}(H \rightarrow WW^*)$ events makes its measurement in the background difficult. Fig. 2.2 depicts a summary of several event cross section measured by ATLAS. It shows the dominating $t\bar{t}$ background with significantly higher a cross section of (829 ± 16) pb compared to $t\bar{t}H$ which reaches (0.670 ± 0.142) pb [45]. Hence, the $t\bar{t}H$ events are three orders of magnitude more rare than $t\bar{t}$ events. Other background processes such as tW , $t\bar{t}Z$ and $t\bar{t}W$ have cross sections higher or close to $t\bar{t}H$.

However, this study focuses on a particular H -boson decay mode. Hence, the branching ratio of the Higgs decay must be taken into account. For the previously stated SM H -boson mass M_H , the theoretical branching ratio of $H \rightarrow WW$ is the second highest at $(25.7 \pm 2.5)\%$ [46]. Furthermore, the W -boson decay ratios are $(67.41 \pm 0.27)\%$ hadronically and $(32.58 \pm 0.47)\%$ [47]. Multiplying the $t\bar{t}H$ cross section with these decay ratios and taking the W -permutations into account yields a final cross section of 0.079 pb for the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ event topology. This results to approximately 10000 expected $t\bar{t}$ events per semileptonic $t\bar{t}(H \rightarrow WW^*)$ event. Taking other background processes into account, semileptonic $t\bar{t}(H \rightarrow WW^*)$ events become even more difficult to measure in background.

2. The Standard Model of Particle Physics

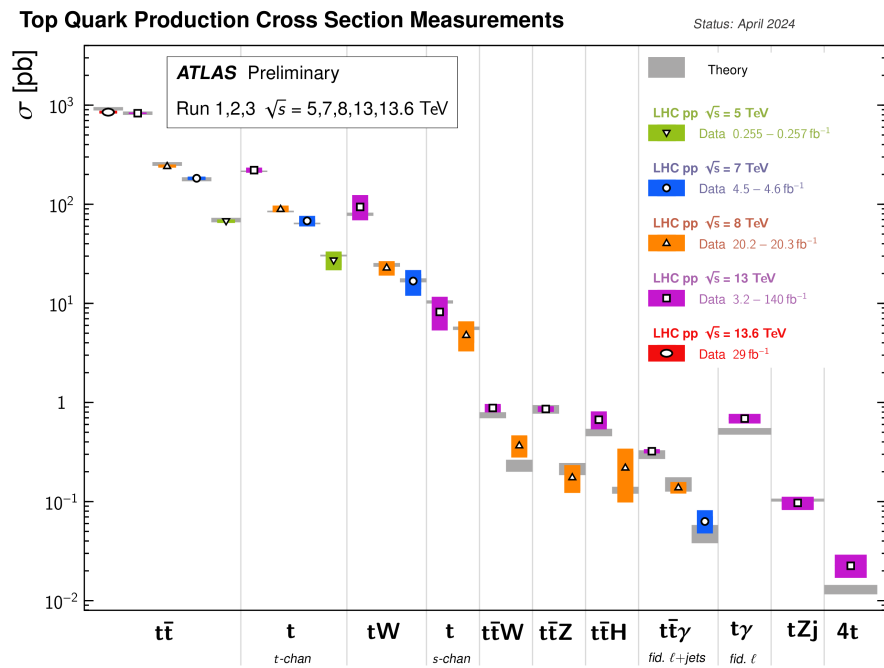


Figure 2.2.: Summary of several t quark related production cross section measurements. The theoretical expectations were calculated at least at NLO [45].

3. Experimental Setup

To collect data on $t\bar{t}(H \rightarrow WW^*)$ events, a high-energy particle collider is necessary. Additionally, a setup for signal detection and suitable reconstruction algorithms are required. This chapter introduces the Large Hadron Collider [48, 49] in Section 3.1 which is used for collecting the data for this project. It is one of the most important particle accelerators worldwide and includes many experiments such as the ATLAS experiment [50]. The ATLAS experiment provides the data for the project and is discussed in great detail in Section 3.2. Other experiments are briefly described in Section 3.3.

3.1. Large Hadron Collider

The Large Hadron Collider (LHC) [48, 49] is the world's largest and most powerful particle accelerator. It's located underground near Geneva, Switzerland, crossing the border between Switzerland and France. Specifically, it's situated at the European Organization for Nuclear Research (CERN). The collider spans 27 km in circumference and reaches depths of up to 175 m. Along the collider, there are several detectors build which will be explored in Section 3.2 and Section 3.3. Figure 3.1 shows an aerial view of the regions above the LHC demonstrating its size and showing the position of the different experiments.

It is used to create and accelerate bunches of protons close to the speed of light. Then a collision of these bunches is induced. During such a collision, partons from the protons interact and give rise to a large multitude of processes. By studying the debris produced by these collisions, physicists can gain insights into fundamental questions about the nature of the universe. Moreover, predictions of the SM introduced in Chapter 2 can be tested with such collisions.

The LHC is operating for certain time periods with long shutdowns in between. The first operation period, Run I, was active from November 2009 until February 2013. It started with a beam energy of 1.2 TeV which was then later increased to 3.5 TeV. In the last year it reached a beam energy of 4 TeV. During the operation, 5.6 fb^{-1} had been accumulated by ATLAS and CMS. Then Long Shutdown 1 followed Run I.

In April 2015, Run II started. The second run reached beam energies of 6.5 TeV and

3. Experimental Setup

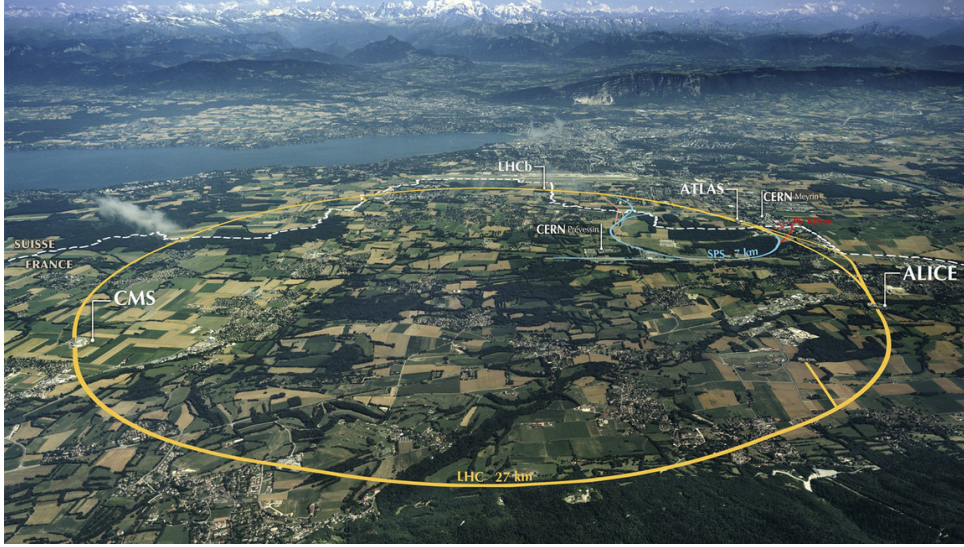


Figure 3.1.: Aerial view above the Large Hadron Collider showing its size. The different experiments are marked along the LHC course. ©CERN

lasted until December 2018. It acquired an integrated production of 160 fb^{-1} . The Long Shutdown 2 then held on until 2022.

The currently active Run III launched in April 2022 with a centre-of-mass energy of 13.6 TeV, 6.8 TeV per beam. It is planned to operate until the coming Long Shutdown 3 planned for 2026.

3.2. The ATLAS Experiment

ATLAS (A Toroidal LHC ApparatuS) [50] is one of the four main experiments at the LHC. Its detector is designed as a general-purpose detector to explore a wide range of physics phenomena [51, 52]. The detector began its operation along the start-up of the LHC. It spans a height of 25 m and a length of 44 m. It is build almost hermetically around the LHC beam pipe as a toroidal shape which can be seen in Figure 3.2. Constructing the detector so close to the interaction point allows measuring the maximum possible phase space. The coordinate system used in ATLAS is important for understanding the detector and is introduced in Section 3.2. Within the detector, there are several layers that are explained in great detail in Section 3.2.1.

Coordinate System for ATLAS

The ATLAS detector is built as a cylinder. Thus, a fitting coordinate system must be defined. Every position in the detector is defined by the z position parallel to the beam

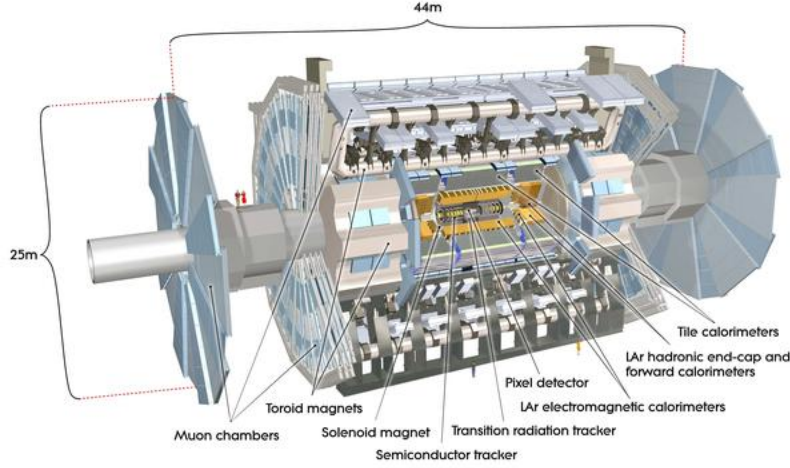


Figure 3.2.: Overview of the ATLAS detector showing its size and structure. Moreover, every important component is labelled. ©CERN

axis, the azimuth angle ϕ along the detector and the pseudo-rapidity η with respect to the collision point of the detector. The latter quantity η is defined by

$$\eta = -\ln \left(\tan \frac{\theta}{2} \right) \quad (3.1)$$

with θ the polar angle. Figure 3.3 illustrates a 2 dimensional projection of the cylinder at a fixed distance from the center.

To properly work with spatial separations in this coordinate system, the variable

$$\Delta R = \sqrt{(\Delta\phi)^2 + (\Delta\eta)^2} \quad (3.2)$$

is introduced. It represents a cone around each point in the coordinate system as shown in Figure 3.3. This should not be confused with R which is typically the radius from the beam pipe. Hence, expressions such as $R\phi$ correspond to the radial scaled azimuth angle and is used in classifying detector resolutions as seen before.

3.2.1. Components of the ATLAS Detector

The detector of the ATLAS experiment is of particular interest for this project. The following sections will cover its elements including the detector modules and magnetic systems in great detail. An overview of the detector is shown in Figure 3.2. The individual layers of the detector are further visualised in Figure 3.4.

3. Experimental Setup

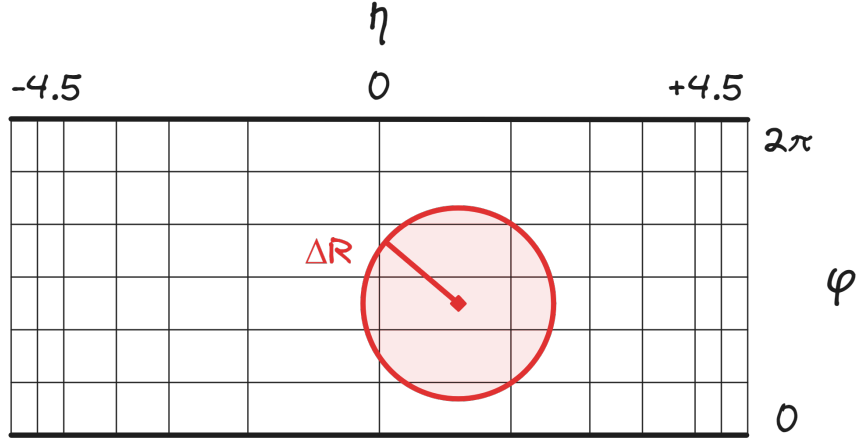


Figure 3.3.: Sketch visualising the ATLAS coordinate system using the azimuth angle ϕ and pseudo-rapidity η . It represents the rolled out detector cylinder. Moreover, it shows how a spatial cone defined by ΔR would look like.

Magnetic System

For the measurement of particle trajectories and their curvature, significant magnetic fields are required. For that reason, ATLAS implemented superconducting magnet system build from Aluminium, Copper and Niobium-Titanium alloy.

It includes a central solenoid magnet surrounding the Inner Detector. It provides a central field of 2 T [51]. Since the central solenoid is placed right before the calorimeter, this magnet is build as thin as possible.

Furthermore, three large air-core toroids generate an additional magnetic field for the outer muon spectrometer. Each consisting of eight symmetrical coils placed radially around the beam axis [51]. These magnet systems are cooled using liquid Helium at 4.5 K [51].

The Inner Detector

The Inner Detector [53, 54] is a combined system of high-resolution detectors close to the beam pipe and continuous tracking elements at outer radii. Its task is to measure the particles' momentum. As mention before, the complete Inner Detector is surrounded by the 2 T central solenoid magnet. Charged particles are deflected inside the detector, due to the Lorentz force. By precisely measuring the curvature, momentum measurements and particle track reconstruction are achieved.

Its detector modules are placed either in the barrel or the end-cap region. The first region is arranged on concentric cylinders surrounding the beam axis and typically covers

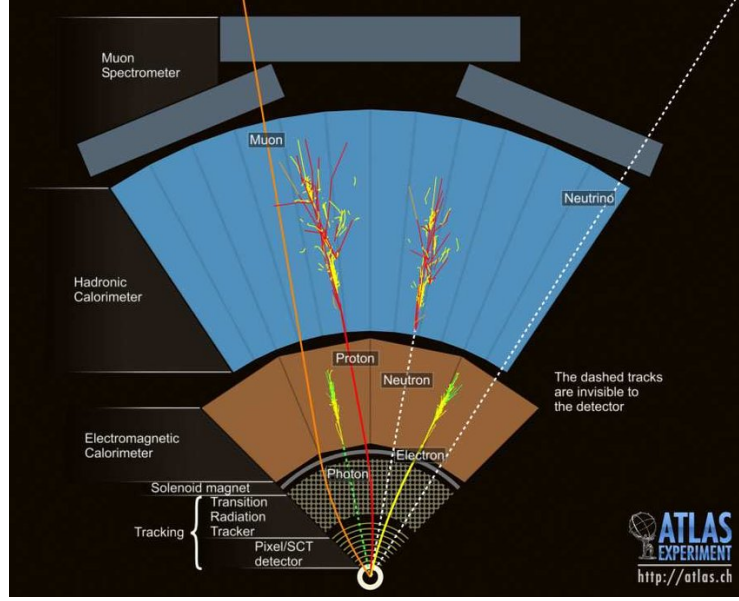


Figure 3.4.: Cross-section of the concentric layers within the ATLAS detector. ©CERN

the low η region. The later regions detectors are installed on disks perpendicular to the beam axis and cover the remaining space for a full coverage up to $\eta \pm 2.5$.

The innermost detector tiles are semiconductor pixel detectors [51, 55]. The first barrel layer is only 4 cm from the beam pipe, covers an area of 0.2 m^2 and measures the complete space up to $\eta = \pm 2.5$. It reaches a resolution of $\sigma_{R\phi} = 12 \text{ }\mu\text{m}$ and $\sigma_z = 66 \text{ }\mu\text{m}$. This layer is followed by two more barrel layers that cover $\eta = \pm 1.7$ and reach the same resolution. To compensate this lower η acceptance, 5 end-cap disks are placed in each side. These cover the remaining space up to $\eta = \pm 2.5$. Each module is only 21.4 mm wide and 62.4 mm long containing 61440 pixels controlled by 16 chips. Every pixel detector has its individual circuit including buffering to store data until the trigger decision. While the precision and radiation resistance of these pixel detectors are remarkable, they also introduce some material before the calorimeters and are expensive. Hence, the amount of pixel detectors that can be included in the detector is limited.

Directly surrounding the pixel modules, the silicon strips are introduced [51, 56]. It is subdivided into four barrel layers for the η -range up to $\eta = \pm 1.4$ and 9 end-cap wheels on each side for the remaining space up to $\eta = \pm 2.5$. Compared to the pixel modules, they achieve slightly lower precisions of $\sigma_{R\phi} = 16 \text{ }\mu\text{m}$ and $\sigma_{z/R} = 580 \text{ }\mu\text{m}$. Each detector is $6.28 \times 6.40 \text{ cm}^2$ in size and includes 768 strips. Moreover, each module consists of four of these silicon detectors.

Straw tube trackers around the previous modules allow for a lot of measured tracking points [51, 57]. In the barrel regions, these are placed parallel to the beam axis. In contrast

3. Experimental Setup

to the end-caps where these are aligned perpendicular to the beam axis. Previously, these were filled with a Xenon gas mixture. However, due some leakage this Xenon was changed to Argon to reduce costs. While these achieve significantly worse resolutions of $\sigma = 170 \mu\text{m}$, they can cover a greater area at lower cost and less material per tracking point. Hence, they strongly contribute the precise momentum measurements.

The Calorimeters

The calorimeters [51, 58] are used to detect charged particles by absorbing them, allowing for a measurement of their deposited energy. The physical process of showering which allow such measurements was previously described in Chapter 2. ATLAS' calorimeter system is subdivided into four parts: electromagnetic calorimeter, hadronic barrel calorimeter, hadronic end-cap calorimeter and hadronic forward calorimeter.

The first of these calorimeters, the electromagnetic calorimeter [51], is built from lead, liquid argon and kapton electrodes. It covers the pseudorapidity range up to $\eta = \pm 3.2$ while the barrel region can detect up to $\eta = \pm 1.475$. It is the innermost calorimeter and is primarily responsible for detecting photons and electrons.

The hadronic calorimeter [51] is build around the electromagnetic calorimeter and is build from plastic scintillator plates and iron absorber in the barrel region and from copper, lead and liquid argon in the end-cap regions. This separation is done to optimise the detector for the varying requirements of the great η range. The barrel region reaches up to $\eta = \pm 1.7$ and the end-caps up to $\eta = \pm 3.2$. Its thickness is one of the most important parameters for the calorimeter, since it has to contain the hadronic showers and thus, minimize punch-through into the adjacent Muon Spectrometer.

The forward hadronic calorimeter [51] is implemented to cover the strongly boosted region of $3.1 < |\eta| = 4.9$. It as well is build from a lead liquid argon mixture with additional electrodes from copper or a tungsten alloy.

The Muon Spectrometer

The outermost measurement part of the ATLAS detector is the Muon Spectrometer [51, 59]. It bends the muon tracks by utilising the previously explained air-core magnets. The magnetic field is mostly orthogonal to the muon trajectories. In the region up to $\eta = \pm 1.0$ the large barrel toroid magnet provides the needed magnetic field. Within the acceptance range of $1.4 \leq |\eta| \leq 2.7$ two smaller end-cap magnets bend the tracks. Between of these regions is the transition region where the deflection of the muon trajectories is done by the magnet field of both magnetic systems. The muon spectrometer consists of four parts.

The first part is the monitored drift tube chamber which make up 800 m^3 in volume. It consists of 30 mm diameter aluminium tubes operating with a mixture of 93% argon and 7% CO_2 . The single-wire resolution is $\sim 80\text{ }\mu\text{m}$. To increase the overall resolution, multilayer pairs of multiple tubes are utilised.

The New Small Wheel project replaced the original Cathode Strip Chambers [59–61]. It was installed after Run II and uses Micro-Mesh Gaseous Structure as well as small-strip thin gap chambers. The former being a detector consisting of a planar electrode and a thin steel mesh forming a drift chamber. It is filled with the same argon- CO_2 mixture as the first drift tube chamber. Its spatial resolution reaches $73\text{ }\mu\text{m}$. The later thin gap chambers consist of a gold-plated tungsten wire grid placed between two electrodes. The electrodes are made from a carbon-epoxy mixture. The detector module reaches a spatial resolution of $100\text{ }\mu\text{m}$ at the rate of up to 20 kHz/cm^2 .

The resistive plate chambers [51, 59, 60] are another gaseous detector. It uses a non-flammable $\text{C}_2\text{H}_2\text{F}_4$ SF_6 gas mixture. Again, the electrodes of the detection chamber are made from carbon. The panels themselves are made polystyrene placed between aluminium sheets. They achieve a spatial resolution of 1 cm and a temporal resolution of 1 ns.

The thin gap chambers [51, 59] are the last chamber of the muon spectrometer. It is a proportional chamber using a flammable gas mixture of 55% CO_2 and 45% n-pentane. Even though it needs additional safety precautions, it is less sensitive to mechanical deformations, yields nearly Gaussian pulse height distributions and has a small dependence on the incident angle. The distance of both electrodes are 2.8 mm.

Trigger System

The aforementioned detector systems are able to detect and record significant amounts of data. The initial bunch-crossing rate of up to 40 MHz result in an interaction rate of $\sim 1\text{ GHz}$ [51] which is too much to save. Approximately only 0.1 kHz of data is suited for permanent storage. Thus, a very fast selection must be defined and implemented that discards most of the events to a manageable level. For that purpose, an online multi-level trigger system is introduced. Each trigger level refines the selection done from the previous level.

The first level (L1) trigger [51, 62] is a hardware-based system and makes the initial selection. It is subdivided into the L1 calorimeter trigger and the L1 muon trigger. The L1 calorimeter trigger utilises signals from the calorimeter in the Cluster and Jet/Energy-sum Processor. The Cluster Processor identifies potential e , γ , τ candidates. This is supplemented by the Jet/Energy-sum Processor which identifies jet candidates and cal-

3. Experimental Setup

culates the total and missing energy. The L1 muon trigger, uses signals from the muon spectrometer to identify μ candidates. The LVL1 decision is formed by the Central Trigger Processor and is based on the signal of the previous two trigger systems as well as some additional detector subsystems. The maximum rate for the L1 trigger is up to 100 kHz, within a decision latency of up to 2.5 μ s. The selected events of the L1 trigger are temporary saved into readout drivers and then readout buffers. Furthermore, the L1 trigger identifies regions of interest, which are further investigated by the second trigger stage.

The event data stays on the readout buffers until the high level (HLT) trigger [51, 62] either rejects or accepts them. The second trigger level is a software-based system and utilises the aforementioned regions of interest. The HLT trigger uses only the necessary information but has access to the full resolution information from the detectors. Its average rate reaches 1.2 kHz which corresponds to approximately 1.2 GB/s data. Its latency is variable but in the order of magnitude 100 μ s. Events that pass both triggers are then transferred to an offline event filter system that performs the final selection.

3.3. Other Experiments

Besides the discussed ATLAS experiment, there are three more main experiments at the LHC: CMS, ALICE and LHCb. This section will briefly mention each of these experiments and compare them to the ATLAS experiment.

The CMS (Compact Muon Solenoid) [63] detector is a general-purpose similar to the ATLAS detector. As its name suggest, a particular focus is set on the precise measurement of high energy muons over a wide range of angles and momenta. Its main goals include muon mass resolution of around 1% at 100 GeV and the ability to determine the charge of muons with momentum $p < 1$ TeV. Other main goals include the reconstruction of charged particles, efficient tagging of τ leptons and b quarks, precise missing transverse energies and isolation of photons and leptons at high luminosities. CMS started its operation with the commissioning of the LHC.

The ALICE (A Large Ion Collider Experiment) [64] focuses on heavy ion collisions. It is designed to measure the physics of strongly interacting matter and the quark-gluon plasma at extreme energies and temperatures. It consists of a central barrel which covers polar angles from 45° to 135° and a forward muon spectrometer covering 2° to 9° . While the ALICE physics program includes the study of light atoms, the main study utilises heavy nuclei Lead-Lead collisions. Its first collisions was recorded in November 2010.

LHCb (Large Hadron Collider beauty) [65] is dedicated to study particles that contain

b quarks, which sometimes are referred as beauty quarks. Unlike the other experiments, its detector is build as a single-arm spectrometer. It covers a forward angle of approximately 0.5° to 17° . The reason lies in LHCb's focus on detecting b hadrons which are predominantly produced in forward and backward cones. The LHCb experiment started the operation at the same time as ATLAS and CMS.

4. Truth Matching

Training a neural network as explained in Chapter 5 requires sufficient training samples. To generate such training samples, a truth matching algorithm is used. The algorithm takes the truth and reconstructed information of MC simulations as inputs, and outputs a jet-parton assignment that can be used for training. The following section gives a high-level explanation of the algorithm.

4.1. Algorithm

To produce the training data, the algorithm begins by scanning the input MC samples. Each sample includes truth information of the partons before interacting with the detector. The truth information also includes the true identification of particles as well as their decay products. Furthermore, the samples include the reconstructed information at detector level. Here, the particles and their properties are simulated as they would appear in data. Pairing the reconstructed and truth information of each event, allows generating training sets that uses the reconstructed information with true labels.

The algorithm analyses each event by iterating through all truth-level final state partons. Around each of these partons, the algorithm searches possible jet matches. All jets within a spatial cone around a parton are considered as its candidate matches. Here, the cone is defined by $\Delta R \leq 0.4$ as introduced in Section 3.2. For the targeted $t\bar{t}(H \rightarrow WW^*)$ semileptonic final state, 8 quarks (3 per t quark + 2 for the hadronic W -boson from the H -boson) have to be matched.

If there is no jet within the cone of a parton, then that parton is not matched. If exactly one candidate is found inside the cone, the jet is saved as the final match for the jet-parton assignment. For cases with more than one candidate match, the jet closest to the particle in ΔR is chosen as the final match.

Each matched jet is made unavailable for other partons. Hence, each jet-parton assignment is unique which is required for training the neural network. Afterwards, the algorithm proceeds to combine the matched particles. The four-vectors of the jets are added up to get the four-vector of the expected resonance particles.

4. *Truth Matching*

Furthermore, several technical optimisations such as hashmaps and bit-shifting operations are implemented to increase the processing speed. To implement bit-shifting, each jet gets an integer number assigned. Each parton corresponds to one of the bits in the integers bit-representation. Bit values of 1 mean the jet is potentially matched to the corresponding parton, while 0 means that the parton is not included within the jets ΔR cone. Each parton has a fixed position in the bit-representation. Hence, checking a potential parton-jet match is done efficiently by bit-operations.

5. SPA-Net: Symmetry Preserving Attention Networks

This chapter introduces the transformer neural network *SPA-Net* [66, 67], which is used for the hadronic jet-parton assignment during event reconstruction. Section 5.1 first explains the classical approach for solving such assignment problems by using the χ^2 minimisation technique. Then Section 5.2 presents the implemented neural network approach.

5.1. Classical χ^2 Method

Traditionally, to determine the jet-parton assignment of an event, a simple χ^2 minimisation approach [68] is used. The method tests every possible combination of jets in the event, calculating the masses m of the resonance particles (t, W, H) and then compares it to the SM prediction M . The method tries to find the best matching mass for all resonance particles. Mathematically, for every assignment the following is calculated

$$\chi^2 = \sum_{i,j} \frac{(m_{q_i} - M_j)^2}{\sigma_j^2}$$

with σ being the SM mass uncertainty. The lowest χ^2 value corresponds to the best matching jets i of each resonance particle j . Thus, it is expected to be the most accurate jet-parton assignment.

This approach is not suited for the targeted event topology of this study. For the targeted $t\bar{t}(H \rightarrow WW^*)$ channel, 8 jets are expected in the semileptonic final state. This leads to high numbers of permutations which slows down the calculation significantly. Even when introducing b -tagging and requiring the method to only swap b -tagged jets with other b -tagged jets, the algorithm would have $6! = 720$ possible light jet permutations per event. Hence, this method is not implemented. Instead, a neural network is trained.

5.2. Modern DNN Method

In this study, the transformer-based deep neural network architecture *SPA-Net* (Symmetry Preserving Attention Networks) [66, 67] is used for predicting the jet-parton assignment. *SPA-Net* was originally developed for fully hadronic $t\bar{t}$ analysis [69]. However, the implemented network is extended to general event topologies [66] and thus, can be used for the targeted $t\bar{t}(H \rightarrow WW^*)$ topology.

Its task is to predict a jet-parton assignment for all jets in the targeted semileptonic final state. This hadronic assignment can then be combined with the leptonic estimation using *Neutrino Weighting* resulting in a full event reconstruction.

One of *SPA-Net*'s features is the utilisation of symmetries in the event topology. These symmetries are implemented in the loss function during training. The loss does not change under predefined symmetrical permutations. Symmetries are classified into two groups: jet and particle symmetries. The jet symmetry describes that labelling of certain particles is symmetric and thus, changing their labels does not change their physical behaviour. In the $t\bar{t}(H \rightarrow WW^*)$ topology, this symmetry is present in the hadronic W -boson decays which produce two quarks $q\bar{q}'$. The particle symmetry depicts the symmetry with respect to the particle origin. Particles that originate from a quark or its respective antiquark are symmetric, when disregarding their charge. Since the charge of a jet is difficult to measure, this symmetry is functional for the event analysis. For the targeted $t\bar{t}(H \rightarrow WW^*)$ topology, the symmetry is applicable in the decays of either the t or $\bar{t}$.

By considering these symmetries, the number of possible permutations is reduced and the processing speed increases. The run-time is reduced to $\mathcal{O}(N_{\text{jets}}^3)$ compared to the baseline χ^2 method with a expected run-time of $\mathcal{O}(N_{\text{jets}}^C)$ [66]. Here $C = 8$ is the number of final state partons or $C' = 6$ when separating b -tagged jets.

SPA-Net's network architecture is visualised in Figure 5.1. It can be subdivided into four components [66]: input embeddings, central transformer, particle transformers and self-attention output layer.

During training, each prediction is split into three steps. First, the amount of valid jet assignment permutations is reduced by splitting up events into sub-structures which are predefined by the event topology. Then each jet-parton assignment sub-problem is solved by applying a ‘Symmetric Tensor Attention’ [66] layer, producing a single tensor. For each sub-assignment an entry is generated including the probability that any particular combination is the correct sub-assignment. These tensors are combined into a final jet-parton prediction by calculating the combined symmetric loss [66]. The approach of splitting up the individual particles allows partial event reconstruction which is vital for this study.

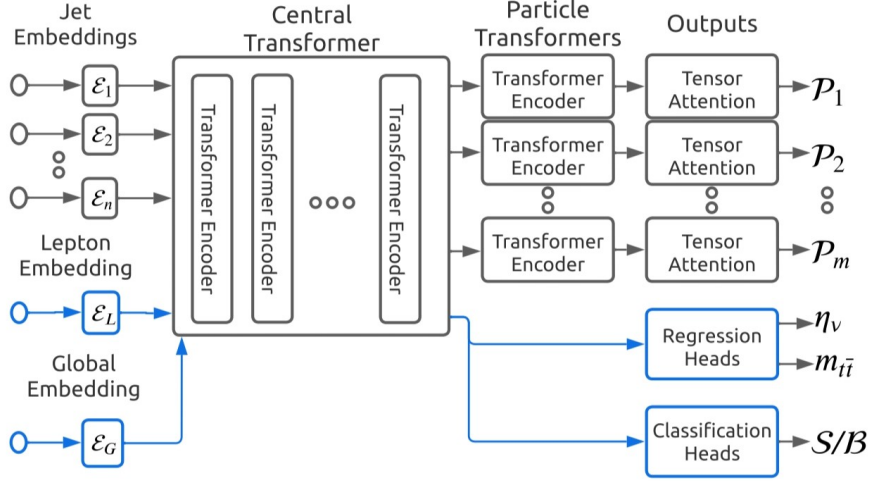


Figure 5.1.: Overview of *SPA-Net*'s architecture including its four regions and the additional embeddings for lepton and global information [67].

Particle symmetries are implemented in the symmetric cross entropy loss during training. Changes which obey these symmetries do not change the loss value. This behaviour is achieved by allowing the network to train any equivalent assignment as long as the overall loss is minimised. Jet symmetries are included in permutation groups which are used by the Symmetric Tensor Attention layer [66].

During inference, the combined symmetric loss is not calculated. Instead, *SPA-Net* iteratively assigns jets to the expected particles by selecting the most likely assignment from each output tensor $\mathcal{P}_p$. The used permutations are bijective to ensure a unique jet-parton assignment. Hence, in cases where one jet is assigned more than once, the highest probability score is selected and the others are evaluated again.

In comparison to other techniques, *SPA-Net* offers several advantages. Firstly, *SPA-Net* is agnostic to the number of input objects. Hence, training and prediction can be done using samples containing any jet multiplicity. Secondly, all predicted jet matches are unique. Furthermore, each prediction is also rated using an assignment probability that represents *SPA-Net*'s confidence for that particular prediction. And lastly, the expected symmetries due to the event topology are predefined which leads to significantly higher processing speed by simplifying the training and prediction task.

For training *SPA-Net* to the targeted topology, a suitable training sample must be created. This necessitates the development of the truth matching algorithm which is explained in Chapter 4.

6. Neutrino Weighting

Neutrino Weighting is a technique to reconstruct events including neutrinos in their final state. It was first implemented by the DØ collaboration in dileptonic $t\bar{t}$ events [70]. The kinematic reconstruction with neutrinos in the final state are underconstrained, due to the unmeasured neutrino particle. Hence, the reconstruction relies on certain assumptions. With these constraints, *Neutrino Weighting* yields an estimation of the unknown parameters by calculating the most likely combination of them.

While the original *Neutrino Weighting* was developed for events with two neutrinos in the final state, this study adapts and modifies the original approach to reconstruct the leptonically decaying W^* -boson in the $H \rightarrow WW^*$ subprocess [71]. For the targeted event topology, the free parameters are the mass, m_{W^*} , of the off-shell W^* -boson and the pseudo-rapidity, η_ν , of the neutrino ν .

The algorithm outputs an estimation for these free parameters and a weight, w , which is in the range of $[0, 1]$. Higher values describe a better agreement between the estimated values and the observed event kinematics. This weight can be used in the event selection to identify the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ topology.

6.1. Parameter Estimation

To apply the *Neutrino Weighting* algorithm, some assumptions need to be made to constrain the otherwise underconstrained $H \rightarrow WW^*$ subsystem. First, the mass of the Higgs boson is assumed to equal the SM prediction, $M_H = 125.20 \text{ GeV}$ [24]. Additionally, the mass of the on-shell W -boson is fixed at $M_W = 80.379 \text{ GeV}$ [24]. Lastly, the on-shell W -boson is further assumed to decay hadronically. Then, the $H \rightarrow WW^*$ subsystem can be reconstructed by estimating the off-shell leptonic W^* -boson and combining it with the hadronic W -boson to form a Higgs boson.

To find the best estimation for W^* , several combinations of the off-shell mass, m_{W^*} , and the pseudo-rapidity of the neutrino η_ν are sampled using a grid-based search. The grid samples 300 η_ν points from $[-3, 3]$ and 250 m_{W^*} points within $[0, 50] \text{ GeV}$. All points are distributed equidistantly.

6. Neutrino Weighting

Due to the assumptions, the $H \rightarrow WW^*$ subsystem is fully constrained [71] and the neutrino transverse momentum p_T^ν can be calculated. This reconstructed neutrino momentum, p_T^ν , can be compared to the observed missing transverse momentum, p_T^{miss} . For that, the transverse momenta are split into their two components p_x and p_y with $p_T^2 = p_x^2 + p_y^2$. The weight of one solution is calculated as

$$w = \exp \frac{(p_x^\nu - p_x^{\text{miss}})^2}{\sigma_x^2} \cdot \exp \frac{(p_y^\nu - p_y^{\text{miss}})^2}{\sigma_y^2}. \quad (6.1)$$

The variable $\sigma_{x/y}$ is the experimental resolution of the missing momentum $p_{x,y}^{\text{miss}}$ and only serves to scale the weight. Thus, it has no physical impact on the result if taken equal in x and y [71]. The implementation sets $\sigma_{x/y} = 10 \text{ GeV}$. The value is chosen since it is the worst resolution at the ATLAS detector for events with transverse energy up to approximately 100 GeV [72]. This is expected to be appropriate for the targeted event topology.

For each event the sampled m_{W^*} and η_ν can be visualised together with the weight of each grid point to form a distribution as seen in Figure 6.1. The maximum of this distribution corresponds to the most likely combination of the unknown parameters. White regions are kinematically forbidden and thus, no solutions can be found for these regions.

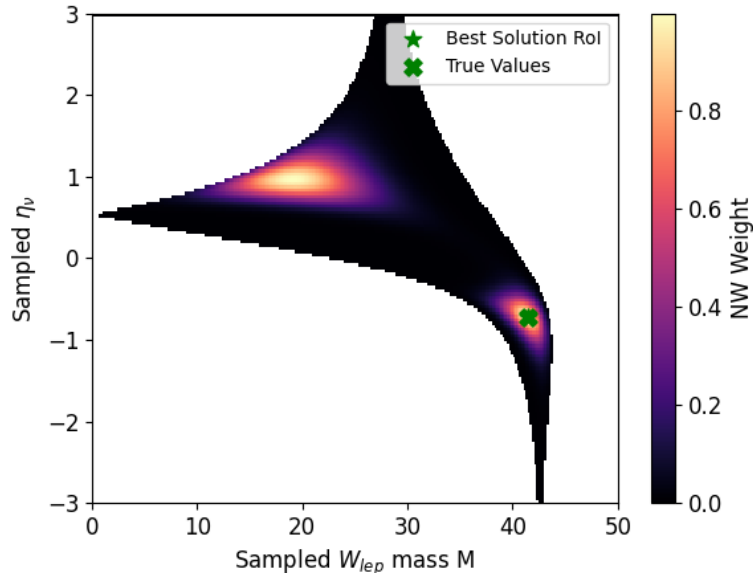


Figure 6.1.: Example of the samples weight distribution for a cherry-picked event. The highest weighted solution (star) and the true values (cross) are marked.

6.2. Optimisation

Since the targeted $t\bar{t}(H \rightarrow WW^*)$ process is rare, it is crucial to maximise the number of reconstructable events. To increase these statistics as well as the accuracy of the estimated values for m_{W^*} and η_ν , two minor optimisations are implemented.

Higgs Mass Smearing

Some events do not yield any solution during the initial grid-based search. Hence, there is no estimation for m_{W^*} or η_ν . To decrease the amount of events with no solution, the assumed Higgs boson mass m_H can be altered slightly. This approach is valid since the on-shell mass definition does not require the exact mass $m_H = M_H$. Rather, it is sufficient that m_H is close to the SM mass M_H for an on-shell Higgs boson.

Hence, if an event has no solution, the assumed Higgs mass m_H is varied in a small range of ± 1 GeV around the SM mass M_H . The grid-based search is repeated for each alternative assumption $m'_H \in [124.20 \text{ GeV}, 126.20 \text{ GeV}]$. The altered Higgs boson mass is sampled in 0.1 GeV steps. If any solution is found using the alternative Higgs mass m'_H , it is used for estimation of η_ν and m_{W^*} .

This additional step increases the processing time for the *Neutrino Weighting* algorithm by 10-50%. The specific time increase is highly dependent on the sample composition. Due to this additional step, around 1% of events that would be rejected become reconstructable.

Regions of Interest

As an additional improvement, the *Neutrino Weighting* algorithm defines regions of interest. These regions are used to refine the estimated solution.

If the algorithm finds at least one solution during the grid-based search, an additional finer grid-based search is started around the highest weighted solution. This second grid is the local region of interest. Within the region of interest, the steps for m_{W^*} and ν_η are more granular to find an even higher weighted, local solution. The final solution is the highest weighted estimation within the region of interest.

The region of interest increase the processing time by 15% independent of the sample composition. However, the average estimation accuracy when using this optimisation is increased by $< 0.1\%$. Hence, this step can be skipped for quick tests and should only be used for final results.

7. Event Selection

This chapter covers the event selection strategy used to separate the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ decays from other background processes. Section 7.1 introduces the event topology in greater detail and explains the deployed assumptions. Section 7.2 defines separation power and showcases some of the investigated variables. The behaviour of several event topologies are compared in respect to these variables. Lastly, sensitive variables are used in Section 7.3 for the final region definition.

7.1. Objects of Target Event Topology

The targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ event topology is introduced in Sec. 2.4. Its final state consists of eight jets and a single lepton-neutrino pair. Six of these jets originate from the $t\bar{t}$ decays, while the remaining two jets stem from the Higgs decay subsystem. Among the eight jets, exactly two are tagged as b -jets which originate from the t quark decay $t \rightarrow Wb$. For the purposes of event reconstruction, three simplifying assumptions are made.

First, it is assumed that the $t\bar{t}$ system decays fully hadronically. This assumption implies that the observed lepton does not come from the t quark decay but is instead produced in the H -boson decay chain. Only then can the implemented *Neutrino Weighting* technique be applied properly.

The second assumption posits that the H -boson is produced on-shell, with a SM mass of $M_H = 125.20 \pm 0.11$ GeV. Given the energy constraints, the Higgs boson cannot decay into two on-shell W -bosons. Therefore, one of the W -bosons must be off-shell.

The third assumption specifies that the W^* -boson decaying leptonically is off-shell, marked by the asterisk *. This is crucial because, without the on-shell constrain on the hadronic W -boson, it would be almost impossible to assign the jets to the W -boson, as any combination of jets could potentially reconstruct an off-shell W^* -boson with an indeterminate mass.

These assumptions constrain the kinematics of the event. Specifically, the energy of the lepton and neutrino is limited to approximately ≤ 50 GeV, caused by the limited

7. Event Selection

off-shell mass of the W^* -boson. Additionally, the two jets assigned to the on-shell W -boson are expected to have an invariant mass close to the SM W -boson mass value of $M_W = 80.3692 \pm 0.0133$ GeV.

As explained in Sec. 2.4, the primary background arises from $t\bar{t}$ events, which have a significantly larger production cross section compared to the signal $t\bar{t}(H \rightarrow WW^*)$ events. $t\bar{t}(H \rightarrow bb)$ and $t\bar{t}(H \rightarrow \tau\tau)$ events also contribute significantly, due to their comparatively high H decay ratio of $(53 \pm 8)\%$ and $(6.0 \pm 0.8)\%$, respectively [46]. Additional backgrounds can arise from $t\bar{t}(H \rightarrow cc)$ events and hadronic or dileptonic $t\bar{t}(H \rightarrow WW^*)$ decays. These background processes are considered during the variable investigation and in the final region definition.

7.2. Separation Power

To compare the performance of using certain variables to separate signal and background, the corresponding distributions are plotted together. Here, signal is defined to only include semileptonic $t\bar{t}(H \rightarrow WW^*)$ events while background includes everything else. Naturally, the background contains the majority of all events. For a fair comparison, both signal and background are normalised to unity and binned using the same steps. The choice of variables is physically motivated by identifying measurable differences in the signal and background processes.

Furthermore, the separation power Γ can be calculated using

$$\Gamma = \frac{1}{2} \sum_i^{\text{bins}} \frac{(s_i - b_i)^2}{s_i + b_i}. \quad (7.1)$$

For each bin i , the variables s_i and b_i equal the normalised signal and background distributions, respectively. Since the separation power Γ is dependent on the binning, the absolute value has no physical meaning. However, when choosing the same binning, the separation power of several variables can be compared to identify the best performing ones.

Jet Multiplicities

Figure 7.1 shows two distributions of events per total jet and b -jet multiplicity, respectively.

As expected, the distribution $t\bar{t}(H \rightarrow bb)$ of events is peaking at higher b -jet multiplicity than any other distribution. While all distributions peak at 2 b -jets, the $t\bar{t}(H \rightarrow bb)$ event

distribution peaks at 3 b -jets due to the two additional b -quarks in the final state. The b -jet separation power $\Gamma = 0.002$ is relatively low, since most backgrounds behave identical to signal in respect to the number of b -jets. Hence, this variable is mainly used to separate $t\bar{t}(H \rightarrow bb)$ background.

The total number of jets shows similar distributions for all events. $t\bar{t}$ events slightly favour low jet multiplicities which can be explained by their final state containing only 6 jets. The targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ final state contains 8 jets. Moreover, hadronic $t\bar{t}(H \rightarrow WW^*)$ events contain 10 jets in their final state. Hence, the number of jets can be used to separate some of the backgrounds. However, due to additional jets or jets lost during reconstruction, the distributions do not clearly peak at their expected jet multiplicities. The separation power $\Gamma = 0.219$ is the highest value of all investigated variables. This supports the importance of this variable for separating signal from background.

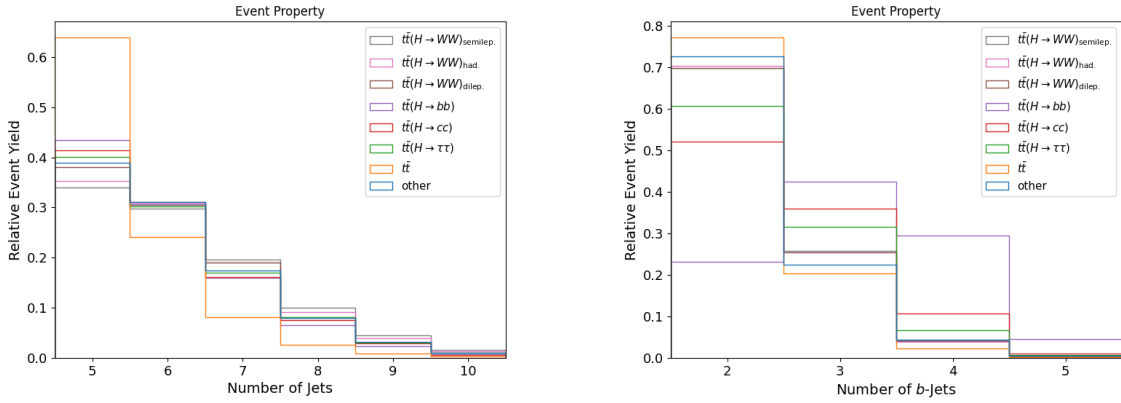


Figure 7.1.: Distributions of different event topologies for (left) the total jet multiplicity and (right) the b -jet multiplicity per event.

Lepton Energies

The distributions in Figure 7.2 show the behaviour of different event topologies for the lepton energies E_{lep} and missing transverse energies E_T^{miss} .

Due to the implemented single lepton triggers, every event contains exactly one lepton and thus, the choice of lepton is unambiguous.

For the missing transverse energy E_T^{miss} , the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ events peak approximately 10 GeV lower than most other processes. In these events, the missing energy is expected to originate from the neutrino decay within the $H \rightarrow WW^*$ subsystem. Hence, E_T^{miss} is limited by the energy of the leptonic W^* -boson decay. Furthermore, the

7. Event Selection

reconstruction in this analysis assumes the leptonic W^* -boson to be off-shell which limits the maximum E_T^{miss} energy to approximately ≤ 50 GeV for an on-shell Higgs boson. Hence, the missing transverse energy can be used to identify events which fulfill the leptonic off-shell W^* -boson assumption. The separation power reaches $\Gamma = 0.030$ which corresponds to a similar sensitivity to signal and background.

The energies of the lepton E_{lep} show a behaviour similar to E_T^{miss} . The targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ events generally have lower lepton energies which again originates from the leptonic off-shell W^* -boson assumption. However, here the dileptonic $t\bar{t}(H \rightarrow WW^*)$ events behave very similar to the semileptonic decays. This similarity is expected since the production of the leptons is identical for both the semileptonic and dileptonic decay mode. Here, the separation power is $\Gamma = 0.012$ which is caused by the very similar behaviour of the dileptonic $t\bar{t}(H \rightarrow WW^*)$ and $t\bar{t}(H \rightarrow \tau\tau)$ backgrounds. The dileptonic $t\bar{t}(H \rightarrow WW^*)$ can be separated by E_T^{miss} since these events contain two neutrinos and thus, generally higher E_T^{miss} .

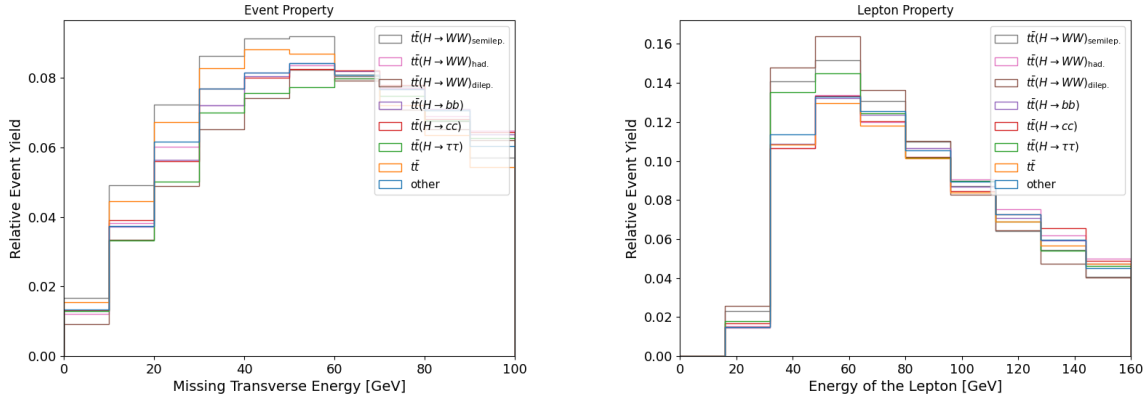


Figure 7.2.: Distributions of different event topologies for (left) the missing energy per event and (right) the energy of the identifies lepton.

SPA-Net Output Probabilities

Figure 7.3 includes two distributions of *SPA-Net*'s output probabilities. These probabilities are calculated during the jet assignment prediction. Events with high probabilities are more likely to be correctly matched. Both the t and the $W_{\text{had.}}$ assignment probabilities show significantly different behaviours for the event topologies.

The t assignment probability generally is higher for the semileptonic and dileptonic $t\bar{t}(H \rightarrow WW^*)$ events which peak at 0.7 and 0.6, respectively. $t\bar{t}$ events score the lowest. Its peak is below 0.05. The 8+ jet limitation causes $t\bar{t}$ events to have at least two additional

jets measured. Hence, the additional possible jet choices reduce the probability of the correct jet assignment. The $t\bar{t}(H \rightarrow bb)$ events also peak at 0.05 which is caused by the two additional b quarks. These b quarks from the H -boson decay can be misidentified as the b quark from the t quark decays. The average of both t_1 and t_2 assignment probability is $\Gamma = 0.212$ which is the second highest separation power of all variables.

Similarly, the $W_{\text{had.}}$ assignment probability also separates the different events topologies. However, the distributions are not as clearly separated compared to the t assignment probabilities. Here, $t\bar{t}$ and $t\bar{t}(H \rightarrow bb)$ peak at around 0.3 while the semileptonic $t\bar{t}(H \rightarrow WW^*)$ peaks at 0.55. This variable reaches a separation power of $\Gamma = 0.090$. Compared to Γ of the t assignment probability, this variable is less sensitive to signal. Still, both output probabilities are used to define the final event regions.

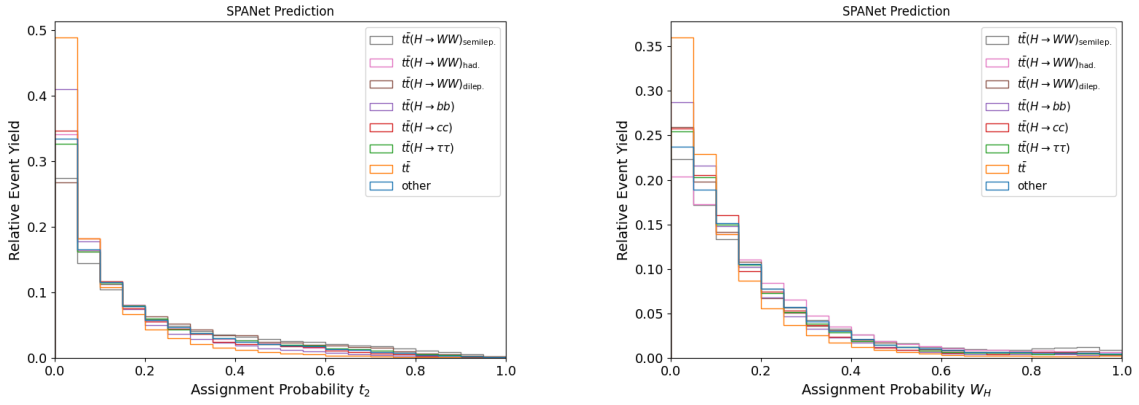


Figure 7.3.: Distributions of different event topologies for *SPA-Net*'s assignment probabilities of (left) the top t_2 and (right) the hadronic W -boson from the Higgs decay.

7.3. Region Definition

To separate signal and background, the overall phase space is subdivided in signal regions S and control regions C . If defined properly, the signal region is expected to include a relatively high purity of the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ events while excluding as much of background contamination as possible. The control regions contain the remaining events with high background yields. Each control region targets one of multiple background processes. Only reconstructable events by the *Neutrino Weighting* algorithm are selected. Given the rarity of the $t\bar{t}(H \rightarrow WW^*)$ process, it is important that the union of all regions covers the entire remaining phase space to avoid discarding events.

7. Event Selection

Table 7.1.: Overview of the defined signal and control regions and their corresponding cuts. Both $C_{t\bar{t}}$ and $C'_{t\bar{t}}$ are included in the same control region.

	S	$C'_{t\bar{t}}$	$C_{t\bar{t}}$	$C_{H \rightarrow bb}$
Total Jet Multiplicity	≥ 8	≥ 8	< 8	—
b -Jet Multiplicity	$= 2$	$= 2$	$= 2$	≥ 3
t Assignment Probability	≥ 0.85	< 0.85	—	—
W_{had} Assignment Probability	≥ 0.85	< 0.85	—	—
Missing Transverse Energy E_T^{miss}	—	—	—	—
Lepton Energy E_l	—	—	—	—

Table 7.2.: Overview of the expected relative yields in each defined regions. The last row summarises the relative proportion of the region in respect to the full MC sampleset.

	S	$C_{t\bar{t}}$	$C_{H \rightarrow bb}$
$t\bar{t}$	99.45 %	99.93 %	99.67 %
$t\bar{t}(H \rightarrow bb)$	0.09 %	0.02 %	0.26 %
$t\bar{t}(H \rightarrow cc)$	0.01 %	0.00 %	0.00 %
$t\bar{t}(H \rightarrow \tau\tau)$	0.02 %	0.01 %	0.01 %
$t\bar{t}(H \rightarrow WW^* \rightarrow qq\bar{q}\bar{q})$	0.08 %	0.01 %	0.01 %
$t\bar{t}(H \rightarrow WW^* \rightarrow l\nu qq)$	0.28 %	0.01 %	0.02 %
$t\bar{t}(H \rightarrow WW^* \rightarrow l\nu l\nu)$	0.01 %	0.00 %	0.00 %
Other	0.04 %	0.01 %	0.01 %
Proportion of Total	0.03 %	77.92 %	22.05 %

The signal and control regions are constructed to be non-overlapping, ensuring that the control regions remain orthogonal to the signal regions. Again, the choice of variables and the selected cuts for each region is driven by their respective separation power. The qualitative cut values are chosen by analysing their separation power. An overview of all regions is given in Tab. 7.1 with their relative yields in Tab. 7.2.

Signal Region

One signal region S is defined to separate as much signal events from background contamination.

The targeted topology expects 8 jets of which two jets are b -tagged. Hence, these variables are crucial for the region definition as summarised in Table 7.1. The total event multiplicity was varied. However, a lower limit decreases the purity significantly due to $t\bar{t}$ events. A higher jet limit on the other hand decreases the total signal events drastically.

Furthermore, the *SPA-Net* assignment probability for each t quark and H -boson are

sensitive to the signal topology. Thus, a cut on these variables is applied which increases the signal purity by a factor of approximately 5. The cut-off value was also varied and the chosen value maximises the signal purity while retaining a reasonable amount of signal events.

As listed in Tab. 7.2, the signal region S consists of 0.28% of signal events which corresponds to 0.53% of all signal events. This signal purity is mainly dominated by $t\bar{t}$ events which have significant higher cross-section and similar kinematics.

Control Regions

Two additional control regions $C_{t\bar{t}}$ and $C_{H\rightarrow bb}$ are defined to separate $t\bar{t}$ and $t\bar{t}(H\rightarrow bb)$ events, respectively.

The $C_{t\bar{t}}$ control region targets the dominating $t\bar{t}$ process. Using the two subregions as defined in Tab. 7.1, 99.93% of events are $t\bar{t}$ events.

For the $C_{H\rightarrow bb}$ control region, the b -jet multiplicity is of particular interest. Since it is the only background process with a higher expected number of b -quarks in the final state, it can be separated by cutting events with more than two b -jets detected. This behaviour can also be seen in Fig. 7.1. This results in a region containing 0.26% $t\bar{t}(H\rightarrow bb)$ events as seen in Tab. 7.2.

8. Results

In this chapter, the results of this study are summarised. First, the validation and performance of the truth matching script are stated in Section 8.1. Sections 8.2 and 8.3 list the results from implementing *SPA-Net* and *Neutrino Weighting*, respectively. The final Section 8.4 includes the final results from combining all analysis steps.

8.1. Truth Matching

This section briefly summarises the validated performance for the truth matching algorithm. While a sufficient truth matching performance is beneficial for creating sufficient training sets, it is not as crucial to optimise for the overall analysis since *SPA-Net* is also able to learn using partial events. Hence, only some optimisations were implemented and tested.

Final Jet Selection

During the matching process, a parton might have more than one potential jet to match. In this case, a metric needs to be defined to decide which jet chosen as the final state. For that reason, final assignment in the case of multiple jet candidates is tested by using three different assignment schemes.

The first matching scheme compares only the spatial separation between the potential jets and the targeted particle. The closest jet when calculating ΔR in respect to the parton is selected as the final match. The second approach utilises only Δp_T of the potential jets and the targeted particle. Here, the closest pair in momentum space is selected as the final match. This scheme is motivated by the CP violating properties of the W -boson which should result in non-symmetric transverse momenta for the decay particles. The last scheme combines the information of both ΔR and Δp_T . For combined scheme, the ΔR was scaled to a range of $[0,4]$ and Δp_t to a range of approximately $[0,10]$. The upper bound one is not exact since there is no p_T limit applied beforehand. Then, the pair that minimises the sum of both variables is chosen as the final match.

8. Results

Table 8.1.: Overview of the three different matching schemes tested for the final jet matching. It shows the success fraction for each resonance parton for $t\bar{t}H$. The highlighted ΔR only approach yields the best results and is chosen for the analysis.

	successful matches in %				
	$t\bar{t}H$	H	$t\bar{t}$	t_1	t_2
ΔR only	1.0	24.0	0.1	8.1	7.8
ΔR & Δp_T	0.9	23.8	0.1	7.9	7.8
Δp_T only	0.9	23.7	0.1	7.8	7.8

Table 8.2.: Overview of the truth matching performance. the first row includes a composition of different $t\bar{t}$ and $t\bar{t}H$ processes. The highlighted second row only includes the targeted semileptonic $t\bar{t}(H \rightarrow WW^*)$ channel. A match is counted as successfully, if all decay particles have a jet matched. If the H does not exist on truth, the event is not counted towards the H or $t\bar{t}H$ column.

	successful matches in %				
	$t\bar{t}H$	H	$t\bar{t}$	t_1	t_2
$t\bar{t}$ & $t\bar{t}H$ composition	1.0	24.0	0.1	8.1	7.8
$t\bar{t}(H \rightarrow WW^*)_{\text{semileptonic}}$	2.0	31.0	9.3	32.3	32.8

The success ratios for matching individual particles or the $t\bar{t}H$ system are summarised in Table 8.1. For the test, 1.5 million events were used and the table shows that the ΔR scheme performs the best. Hence, it is chosen for this analysis.

Performance by Process

To further investigate the performance of the truth matching algorithm, its overall performance is analysed. Using a semileptonic $t\bar{t}(H \rightarrow WW^*)$ and a composition of different $t\bar{t} + t\bar{t}H$ MC samples, the number of successful events were counted. A particle match is successful when the particle exists on truth and the matching scripts finds a jet for each of its decay particles. The combined $t\bar{t}H$ column is defined as successful if both t -quarks and the H -boson are successfully matched.

The summary of the overall performance can be found in Table 8.2. The truth matching algorithms performs better on the semileptonic $t\bar{t}(H \rightarrow WW^*)$ subset. However, this discrepancy and overall low performance for the full event matches is not problematic. While the output contains less assignments for training, *SPA-Net* is able to learn with partial events. Hence, all events that include some matched jet are kept to use in training.

Table 8.3.: Summary of the *SPA-Net*'s final architecture and hyperparameters. Parameters marked with * are not optimised.

Epochs	400 (early stop at 122)
Batch Size	2048
Training / Evaluation Split*	80 % / 20 %
Learning Rate	0.0015
Hidden Layers	16
Transformer Layers	64
Initial Embedding Layers	16
Position Embedding Layers	16
Include Skip Connection	True
Use Partial Events	True

8.2. SPA-Net

This section covers the jet-parton assignment using *SPA-Net*. The final neural network is optimised to correctly assign jet for the event reconstruction. Additional studies focusing on events with wrong jet assignments were conducted.

Final Model

To train *SPA-Net*, several training sets were used and evaluated. The final model is trained on a MC set containing 2.5 million MC events composed of $t\bar{t}$ and several $t\bar{t}H$ events with additional $t\bar{t}(H \rightarrow WW^*)$ events injected to improve the network performance. All events are preselected to have exactly one lepton and at least five detected jets.

Moreover, the architecture and other hyperparameters were optimised in a grid-search to find the best performing network. The final choice of parameters is listed in Tab. 8.3. The following results are based on this final model.

Assignment Matrix

To better understand *SPA-Net*'s output, the predicted assignment for each jet is plotted vs the true labels from the truth matching algorithm. The resulting assignment matrix is shown in Figure 8.1. Note that each column is normalised to unity. Objects that have no truth matched jet are not considered.

A diagonal matrix would correspond to a perfect prediction. The (q_1, q_2) 2x2 submatrices for t_1, t_1 and $W_{\text{had.}}$ show the predefined allowed permutations during learning. Hence, a off-diagonal term within these submatrices is considered as a correct assignment.

8. Results

Table 8.4.: Overview of *SPA-Net*'s particle reconstruction performance. Each particle is counted as correctly assigned when all daughter particles are correctly matched. Swaps within the allowed permutations are also considered as correctly matched. If any particle has no true assignment, it is not counted. The event is considered as correctly assigned if both the $t\bar{t}$ and hadronic W from the H -boson are correctly matched. Thus, this is only possible for $t\bar{t}H$ events.

	correct assignments in %				
	t_1	t_2	$t\bar{t}$	$W_{\text{had.}}$	event
$t\bar{t}(H \rightarrow WW^*)_{\text{semileptonic}}$	XX.X	XX.X	XX.X	XX.X	XX.X
$t\bar{t}(H \rightarrow WW^*)_{\text{hadronic}}$	XX.X	XX.X	XX.X	XX.X	XX.X
$t\bar{t}(H \rightarrow b\bar{b})$	XX.X	XX.X	XX.X	-	-
$t\bar{t}(H \rightarrow \tau\tau)$	XX.X	XX.X	XX.X	-	-
$t\bar{t}$	XX.X	XX.X	XX.X	-	-

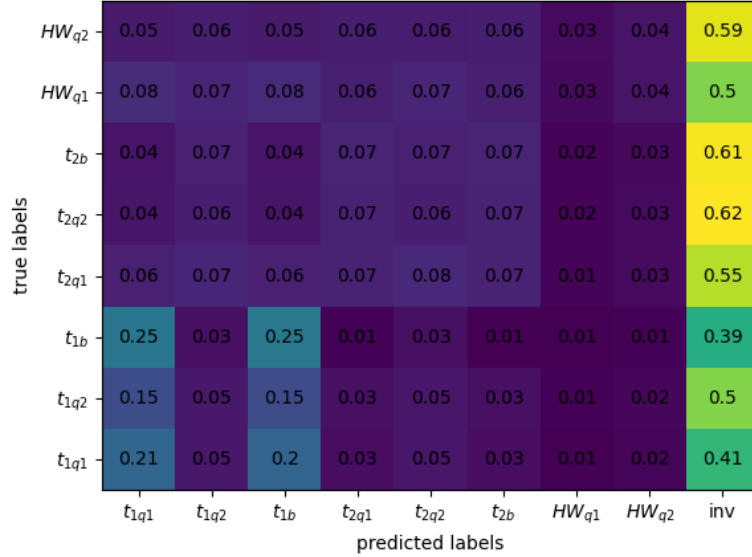


Figure 8.1.: Overview of *SPA-Net* assignment matrix showing the predicted and true labels. Each column is normalised to unity. The predefined permutations between the untagged light jets are clearly visible.

Assignment Efficiencies

The evaluation of *SPA-Net*'s overall assignment is done for each resonance particle (t_1 , t_2 , $W_{\text{had.}}$) and is further split into different processes. The results are listed in Table 8.4. These values are based on the final MC run including approximately 1.5 million events.

Probability Output

Furthermore, the used *SPA-Net* implementation outputs a detection and assignment probability for each resonance particle ($t_1, t_2, W_{\text{had.}}$). These probabilities describe *SPA-Net*'s confidence, whether the targeted resonance particle exists and whether it is assigned correctly. Additionally, a marginal probability is calculated as the product of assignment and detection probability. These probabilities are visualised in Figure 8.2.

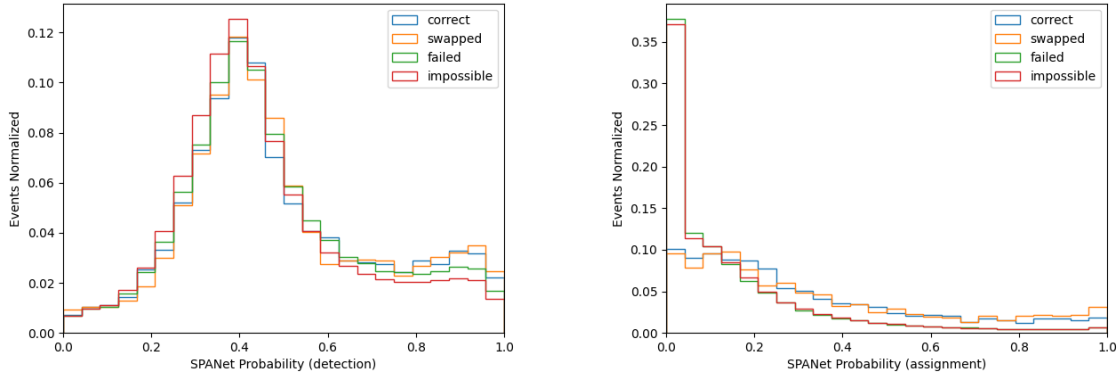


Figure 8.2.: Normalised distribution for *SPA-Net*'s detection (left) and assignment (right) probability. The distributions are split by correct, incorrect and invalid events. Invalid events correspond to events where the targeted resonance does not exist. Swapped assignments correspond to swapped assignments within the predefined permutations.

8.3. Neutrino Weighting

Validation

To validate the accuracy of the NW algorithm, the estimated solutions are compared to the true values of the MC samples. Here, the *Neutrino Weighting* algorithm also uses the truth information as input. Hence, effects due to simulated detector inefficiencies are ignored in the comparison.

Higgs Mass

As an additional validation step, the assumed Higgs mass m_H is varied. The used MC samples for validation uses a true mass of $M_H = 125 \text{ GeV}$. The *Neutrino Weighting* algorithm is tested by assuming the correct mass of $m_H = M_H$ and a slightly off-set mass

8. Results

$m_H = 125.20 \text{ GeV}$.

Fig. 8.3 shows the difference of the estimated values of η_ν and m_W for both assumptions. The heatmaps clearly show the sensibility of the algorithm in respect to the assumed Higgs mass. Furthermore, for a correct assumption $m_H = M_H$, the distribution peaks around $(\nu_\eta = 0, m_W = 0 \text{ GeV})$ as expected. Using a different Higgs mass significantly changes the estimated mass its decay particle W_{lep} . Hence, when using $m_H = 125.20 \text{ GeV}$ the estimated mass m_W is increased by around 200 MeV in respect to the true value due to kinematic conservation laws.

Both observations behave as physically expected, indicating the *Neutrino Weighting* algorithm is working properly.

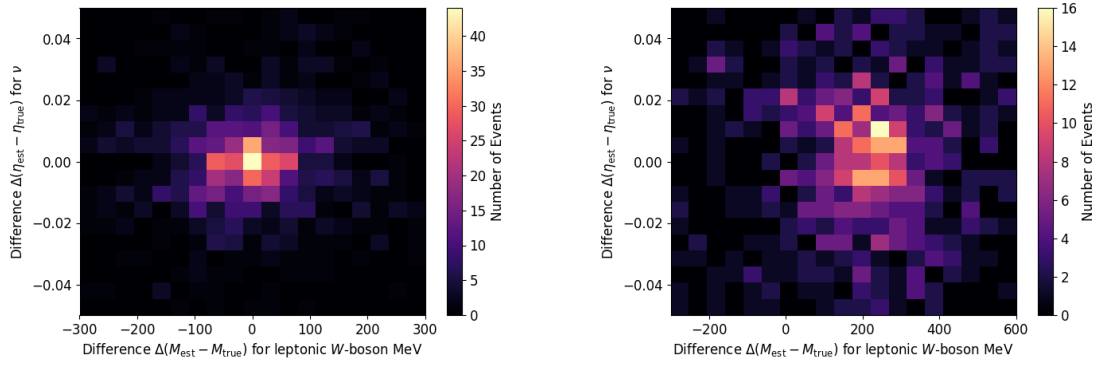


Figure 8.3.: Comparison of two heatmaps showing the difference between the estimated and true values. The two heatmaps use (left) the correct $m_H = 125 \text{ GeV}$ and (right) the offset $m_H = 125.20 \text{ GeV}$.

Performance

8.4. Combined Data Run

Regions

Reconstruction

9. Discussion

9.1. Summary

9.2. Future Aspects

This section outlines potential tasks for the future. While this study has provides insights into the $t\bar{t}(H \rightarrow WW^*)$ event separation, there remain several opportunities for refinement in both the *SPA-Net* training and *Neutrino Weighting* algorithm. These potential future tasks are explained briefly. However, due to time constraints these optimisations exceeded the scope of this work.

Neutrino Weighting Optimisation

Retain SPA-Net without Single Lepton Triggers

Include Systematics

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A. Temporary Other Stuff

A. Temporary Other Stuff

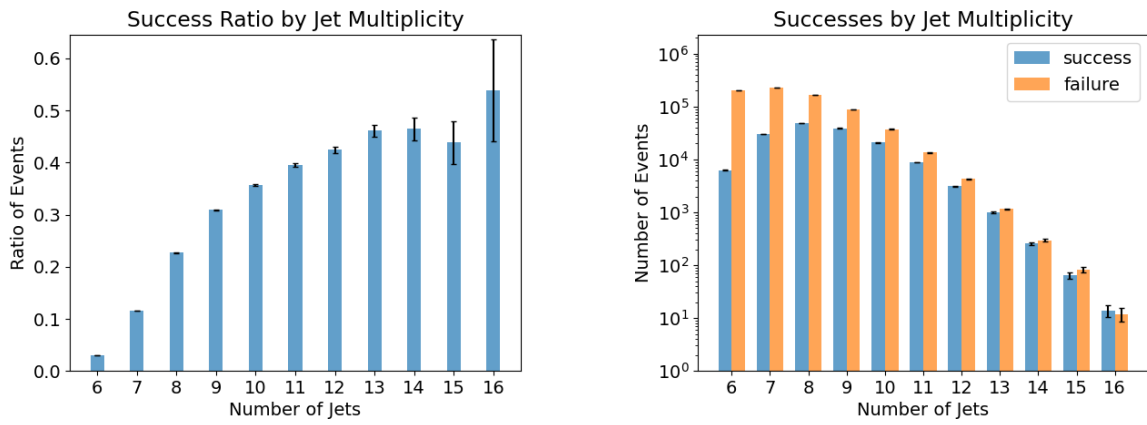


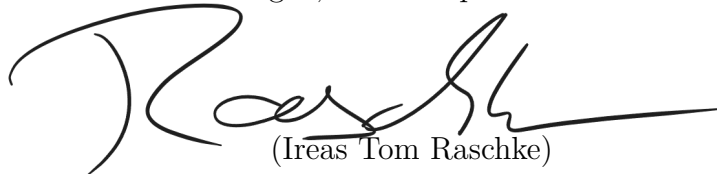
Figure A.1.: Overview of the total successful event count (left) and relative success rate (right) for the matching algorithm subdivided by the number of jets in the event.

B. Declaration

Erklärung nach §13(9) der Prüfungsordnung für den Bachelor-Studiengang Physik und den Master-Studiengang Physik an der Universität Göttingen: Hiermit erkläre ich, dass ich diese Abschlussarbeit selbständig verfasst habe, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe und alle Stellen, die wörtlich oder sinngemäß aus veröffentlichten Schriften entnommen wurden, als solche kenntlich gemacht habe.

Darüberhinaus erkläre ich, dass diese Abschlussarbeit nicht, auch nicht auszugsweise, im Rahmen einer nichtbestanden Prüfung an dieser oder einer anderen Hochschule eingereicht wurde.

Göttingen, den 8. September 2024



(Ireas Tom Raschke)